

A proof of the generating function for OEIS A199085

Third derivatives of parenthesized self-exponentials

A self-contained mathematical article and computational companion

September 20, 2026

Abstract

Let $a(n)$ count the distinct third derivatives at $x = 1$ of fully parenthesized exponentiation expressions containing n copies of x . We prove

$$a(1) = a(2) = 1, \quad a(n) = \left\lfloor \frac{n^2 - 2}{3} \right\rfloor \quad (n \geq 3),$$

and hence the rational generating function conjectured in OEIS A199085. The proof does not extrapolate a recurrence from computed terms. Instead, it classifies all possible pairs of second and third derivatives. At fixed leftmost nesting depth k , the normalized third derivatives form the arithmetic progression $k^2 + 2, k^2 + 4, \dots, k^2 + 2(n - k - 1)$, apart from one extremal singleton. Only progressions of the same parity can overlap, and their ordered endpoints permit exact counting without an inclusion–exclusion expansion. A constructive realization proves that no values in these progressions are missing. We also give recurrences, extremal values, worked examples, and independently implemented exact-arithmetic checks.

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1 The problem and the result

A fully parenthesized exponentiation expression is defined recursively: x is an expression, and if G and H are expressions, then G^H is an expression. Its size is the number of occurrences of x . Let $\mathcal{E}_n$ denote the expressions of size n . They are interpreted as positive real-analytic functions on $(0, \infty)$, with

$$(G^H)(x) = \exp(H(x) \log G(x)).$$

Thus the derivative at 1 has no branch ambiguity. We count values, not expressions: two different parenthesizations that give the same third derivative contribute only one value.

Define

$$\mathcal{D}_n = \{F'''(1) : F \in \mathcal{E}_n\}, \quad a(n) = |\mathcal{D}_n|, \quad A(z) = \sum_{n \geq 1} a(n)z^n. \quad (1.1)$$

The generating-function variable is z , to distinguish it from the variable x in the functions being differentiated.

The OEIS entry attributes the sequence to Vladimir Reshetnikov and the conjectured generating function to Alois P. Heinz, dated November 2, 2011. A program credited to Vladimir Joseph Stephan Orlovsky, dated February 1, 2012, evaluates the quadratic formula proved below [1]. The entry was consulted on September 20, 2026; its generating function was still labeled conjectural there. This article supplies a proof from the defining expressions, without making a separate claim of bibliographic priority.

Theorem 1.1 (The generating function). *For the sequence in (1.1),*

$$a(n) = \begin{cases} 1, & n = 1, 2, \\ \left\lfloor \frac{n^2 - 2}{3} \right\rfloor, & n \geq 3. \end{cases} \quad (1.2)$$

Its ordinary generating function is

$$A(z) = \frac{z}{1-z} + \frac{z^3}{(1-z)^3} - \frac{z^7}{(1-z)^2(1-z^3)} \quad (1.3)$$

$$= \frac{z(1-z+z^2+2z^4-z^6)}{(1-z)^2(1-z^3)} \quad (1.4)$$

$$= \frac{z(z^6 - 2z^4 - z^2 + z - 1)}{z^5 - 2z^4 + z^3 - z^2 + 2z - 1}. \quad (1.5)$$

The last expression is exactly the proposed OEIS generating function, with its variable renamed to z .

The substantive step is not the rational-function simplification. It is determining the complete set of derivative values and then counting its overlaps. We prove a stronger classification in [theorem 4.2](#); [sections 6 and 7](#) then prove [theorem 1.1](#).

2 The analytic reduction to two integer statistics

Every expression F satisfies

$$F(1) = 1, \quad F'(1) = 1. \quad (2.1)$$

Indeed these identities hold for x . If they hold for G and H , then $G^H(1) = 1$ and

$$(G^H)' = G^H \left(H' \log G + H \frac{G'}{G} \right),$$

which gives $(G^H)'(1) = 1$. Structural induction proves (2.1).

Lemma 2.1 (Derivative composition law). *Write*

$$p(F) = F''(1), \quad q(F) = F'''(1).$$

For $F = G^H$,

$$p(F) = p(G) + 2, \tag{2.2}$$

$$q(F) = q(G) + 3p(G) + 3p(H) + 3. \tag{2.3}$$

Proof. Set $L = \log G$ and $V = HL$, so that $F = e^V$. Using (2.1), at $x = 1$ we obtain

$$L = 0, \quad L' = 1, \quad L'' = p(G) - 1, \quad L''' = q(G) - 3p(G) + 2.$$

Consequently,

$$\begin{aligned} V(1) &= 0, & V'(1) &= 1, \\ V''(1) &= 2H'(1)L'(1) + H(1)L''(1) = p(G) + 1, \\ V'''(1) &= 3p(H) + 3(p(G) - 1) + q(G) - 3p(G) + 2 \\ &= q(G) + 3p(H) - 1. \end{aligned}$$

Finally,

$$(e^V)'' = e^V (V'' + (V')^2), \quad (e^V)''' = e^V (V''' + 3V'V'' + (V')^3),$$

which yield (2.2) and (2.3). □

Introduce the normalized statistics

$$k(F) = \frac{F''(1)}{2}, \quad b(F) = \frac{F'''(1)}{3}. \tag{2.4}$$

Since $k(x) = b(x) = 0$, the composition law becomes

$$\boxed{k(G^H) = k(G) + 1, \quad b(G^H) = b(G) + 2k(G) + 2k(H) + 1.} \tag{2.5}$$

It follows by induction that $k(F)$ and $b(F)$ are nonnegative integers. In particular, every third derivative in question is divisible by 3. Multiplication by 3 is injective, so counting the normalized values $b(F)$ is equivalent to counting the actual derivatives.

A useful asymmetry is already visible: $b(H)$ does not occur in (2.5). The exponent contributes its second derivative, but not its third derivative. This is the analytic reason why two statistics suffice.

3 The left spine and its leaf budget

View an expression as a rooted ordered binary tree: an internal node represents exponentiation, its left subtree is the base, and its right subtree is the exponent. The *left spine* is the path obtained by repeatedly following the left child from the root to a leaf. Its length is the number of internal nodes on that path.

The first identity in (2.5) says precisely that $k(F)$ is the length of the left spine. Equivalently, every non-leaf expression has a unique decomposition

$$F = (\cdots ((x^{H_1})^{H_2}) \cdots)^{H_k}, \quad k = k(F), \quad (3.1)$$

where H_1 is the exponent nearest the bottom of the spine. If m_i is the size of H_i , then

$$n = 1 + \sum_{i=1}^k m_i, \quad m_i \geq 1. \quad (3.2)$$

The single extra leaf is the bottom leftmost x .

Lemma 3.1 (Spine formula). *For (3.1), let $s = \sum_{i=1}^k k(H_i)$. Then*

$$F''(1) = 2k, \quad F'''(1) = 3k^2 + 6s. \quad (3.3)$$

Proof. After i nodes of the spine have been assembled, the left-depth statistic is i . The i th use of (2.5) therefore adds $2(i-1) + 2k(H_i) + 1$ to b . Starting from $b(x) = 0$ gives

$$b(F) = \sum_{i=1}^k (2i-1) + 2 \sum_{i=1}^k k(H_i) = k^2 + 2s.$$

The definition of $k(F)$ gives the second derivative as well. □

Each H_i of size m_i has at most $m_i - 1$ internal nodes, and hence $0 \leq k(H_i) \leq m_i - 1$. Moreover, $k(H_i) = 0$ holds exactly when H_i is a single leaf. Thus

$$0 \leq s \leq \sum_{i=1}^k (m_i - 1) = n - 1 - k. \quad (3.4)$$

There is an important boundary distinction. If $k = n - 1$, every $m_i = 1$ and $s = 0$. If $k \leq n - 2$, at least one H_i is not a leaf, so $s \geq 1$. This explains both the missing value k^2 in a non-extremal slice and the exceptional singleton at maximal depth.

4 Every admissible pair is realized

The bounds just obtained do not alone count the values: one must prove that all intermediate s occur. The following simple construction makes that step explicit.

Let R_r be the right-associated comb on r leaves, defined by $R_1 = x$ and $R_{r+1} = x^{R_r}$. Let C_r be the left-associated comb, defined by $C_1 = x$ and $C_{r+1} = (C_r)^x$.

Lemma 4.1 (Prescribed left depth). *For every $m \geq 2$ and $1 \leq h \leq m - 1$, an expression of size m with left depth h exists.*

Proof. Start from $x^{R_{m-h}}$, and apply $h - 1$ additional operations $Y \mapsto Y^x$. The resulting expression has

$$1 + (m - h) + (h - 1) = m$$

leaves and left depth $1 + (h - 1) = h$. All indicated indices are valid because $m - h \geq 1$. $\square$

Theorem 4.2 (Complete derivative-state classification). *Let*

$$\mathcal{S}_n = \{(k(F), b(F)) : F \in \mathcal{E}_n\}.$$

Then $\mathcal{S}_1 = \{(0, 0)\}$, and for every $n \geq 2$,

$$\mathcal{S}_n = \{(n - 1, (n - 1)^2)\} \cup \bigcup_{k=1}^{n-2} \{(k, k^2 + 2s) : 1 \leq s \leq n - k - 1\}. \quad (4.1)$$

In particular,

$$\mathcal{D}_n = \{3(n - 1)^2\} \cup \bigcup_{k=1}^{n-2} \{3k^2 + 6s : 1 \leq s \leq n - k - 1\} \quad (n \geq 2). \quad (4.2)$$

Proof. Necessity follows from [lemma 3.1](#) and (3.4), together with the distinction between $k = n - 1$ and $k \leq n - 2$.

For sufficiency, the extremal pair is realized by C_n . Now fix $1 \leq k \leq n - 2$ and $1 \leq s \leq n - k - 1$. Choose H_1 of size $n - k$ and left depth s , using [lemma 4.1](#), and put $H_2 = \cdots = H_k = x$. The expression in (3.1) then has

$$1 + (n - k) + (k - 1) = n$$

leaves, left depth k , and $\sum_i k(H_i) = s$. Its normalized third derivative is $k^2 + 2s$. This realizes every pair on the right-hand side of (4.1). $\square$

Corollary 4.3 (Number of joint states). *For every $n \geq 1$,*

$$|\mathcal{S}_n| = 1 + \binom{n - 1}{2}. \quad (4.3)$$

Proof. For $n \geq 2$, the different k -slices in (4.1) have distinct first coordinates, so their sizes add:

$$|\mathcal{S}_n| = 1 + \sum_{k=1}^{n-2} (n - k - 1) = 1 + \frac{(n - 1)(n - 2)}{2}.$$

For $n = 1$ the same formula is 1. $\square$

Remark 4.4. Equation (4.3) does not yet give $a(n)$. Different second derivatives can accompany the same third derivative. The first such collision occurs at $n = 7$. The next section counts these collisions exactly.

5 Counting the overlapping progressions

For fixed n and $1 \leq k \leq n-2$, define

$$P_k = \{k^2 + 2s : 1 \leq s \leq n - k - 1\}. \quad (5.1)$$

This is a step-2 integer interval with lower and upper endpoints

$$L_k = k^2 + 2, \quad U_k = k^2 + 2n - 2k - 2 = (k-1)^2 + 2n - 3, \quad (5.2)$$

and cardinality $|P_k| = n - k - 1$. The possible normalized third derivatives are their union together with $\{(n-1)^2\}$.

The singleton is disjoint from all the progressions. For $n \geq 3$, U_k is increasing in k , and

$$(n-1)^2 - U_{n-2} = (n-1)^2 - ((n-2)^2 + 2) = 2n - 5 > 0.$$

For $n = 2$ there are no progressions.

Next, P_k consists of integers congruent to k modulo 2. Thus even-indexed and odd-indexed progressions are disjoint. Within either parity, both endpoints strictly increase with k .

Lemma 5.1 (Only the preceding same-parity interval matters). *For $3 \leq k \leq n-2$,*

$$\left| P_k \cap \bigcup_{j < k} P_j \right| = |P_k \cap P_{k-2}| = (n - 3k + 3)_+, \quad (5.3)$$

where $(u)_+ = \max(u, 0)$.

Proof. Different parities cannot intersect. Among earlier progressions of the same parity, P_{k-2} has the largest upper endpoint. If a value $v \in P_k$ lies in any earlier P_j of the same parity, then

$$L_{k-2} < L_k \leq v \leq U_j \leq U_{k-2}.$$

Its parity is that of P_{k-2} , so $v \in P_{k-2}$. This proves the equality of the two intersections.

Since $U_{k-2} < U_k$ and $L_{k-2} < L_k$, a nonempty intersection runs from L_k to U_{k-2} in steps of 2. Its cardinality is

$$\begin{aligned} \frac{U_{k-2} - L_k}{2} + 1 &= \frac{(k-2)^2 + 2n - 2(k-2) - 2 - (k^2 + 2)}{2} + 1 \\ &= n - 3k + 3. \end{aligned}$$

Taking the positive part also covers the empty case. □

This lemma is what makes triple and higher intersections harmless. We add the progressions one at a time and subtract only those values already present. A value occurring in several earlier progressions is still subtracted just once when the next progression is inserted. There is no assumption that pairwise overlaps are mutually disjoint.

Proposition 5.2 (Exact overlap formula). *For every $n \geq 1$,*

$$a(n) = 1 + \binom{n-1}{2} - \sum_{j \geq 2} (n-3j)_+. \quad (5.4)$$

The sum is finite: only $j \leq \lfloor (n-1)/3 \rfloor$ contribute.

Proof. For $n \geq 3$, sum the sizes of the progressions, subtract the previous union intersection from [lemma 5.1](#) each time, and then add the disjoint singleton:

$$a(n) = 1 + \sum_{k=1}^{n-2} (n-k-1) - \sum_{k=3}^{n-2} (n-3k+3)_+.$$

In the last sum put $j = k-1$. All subsequent terms are zero, so the upper limit can be replaced by infinity. The first sum is $\binom{n-1}{2}$. For $n = 1, 2$, the right-hand side of (5.4) is 1, agreeing with the direct calculation. $\square$

The factor 3 in the overlap cutoff is structural. For consecutive same-parity slices, the endpoint separation contains $3k - n$. This, rather than any empirical fit, is the origin of the eventual period 3.

6 Deriving the conjectured generating function

We now sum (5.4). All calculations can be interpreted in the ring of formal power series. In particular, exchanging the two sums below is justified coefficient by coefficient: at a fixed degree, only finitely many terms are nonzero.

The first two contributions are

$$\sum_{n \geq 1} \left(1 + \binom{n-1}{2} \right) z^n = \frac{z}{1-z} + \frac{z^3}{(1-z)^3}. \quad (6.1)$$

For the correction, fix $j \geq 2$. Substituting $n = 3j + r$ gives

$$\sum_{n \geq 1} (n-3j)_+ z^n = z^{3j} \sum_{r \geq 1} r z^r = \frac{z^{3j+1}}{(1-z)^2}.$$

Therefore

$$\sum_{n \geq 1} \sum_{j \geq 2} (n-3j)_+ z^n = \frac{1}{(1-z)^2} \sum_{j \geq 2} z^{3j+1} = \frac{z^7}{(1-z)^2(1-z^3)}. \quad (6.2)$$

Subtracting (6.2) from (6.1) proves (1.3). Notice that the correction begins at z^7 , exactly where collisions between different depths first appear.

For completeness, the algebraic conversion to the OEIS form is

$$\begin{aligned} A(z) &= \frac{(z - 2z^2 + 2z^3)(1 + z + z^2) - z^7}{(1-z)^3(1+z+z^2)} \\ &= \frac{z - z^2 + z^3 + 2z^5 - z^7}{1 - 2z + z^2 - z^3 + 2z^4 - z^5} \\ &= \frac{z(z^6 - 2z^4 - z^2 + z - 1)}{z^5 - 2z^4 + z^3 - z^2 + 2z - 1}. \end{aligned}$$

The final line multiplies numerator and denominator by -1 . Thus the stated rational generating function follows directly from an exact count of all defining derivative values.

7 The closed formula and its three residue classes

We also evaluate (5.4) explicitly. For $n \geq 4$, put

$$q = \left\lfloor \frac{n-1}{3} \right\rfloor \geq 1.$$

Then

$$\sum_{j \geq 2} (n-3j)_+ = \sum_{j=2}^q (n-3j) = (q-1)n - 3 \left(\frac{q(q+1)}{2} - 1 \right).$$

This identity includes the empty sum at $q = 1$. Substitution gives

$$a(n) = \frac{n^2 - n - 2qn + 3q^2 + 3q - 2}{2}. \quad (7.1)$$

Writing n by its residue modulo 3 yields

$$\begin{aligned} a(3m) &= 3m^2 - 1, & m \geq 1, \\ a(3m+1) &= 3m^2 + 2m - 1, & m \geq 1, \\ a(3m+2) &= 3m^2 + 4m, & m \geq 1. \end{aligned} \quad (7.2)$$

For the first line at $m = 1$, $a(3) = 2$ is checked directly; all other cases follow from (7.1). Because the only square residues modulo 3 are 0 and 1, (7.2) is precisely

$$a(n) = \left\lfloor \frac{n^2 - 2}{3} \right\rfloor \quad (n \geq 3).$$

Together with $a(1) = a(2) = 1$, this finishes the proof of [theorem 1.1](#).

7.1 Why the first two terms must be separated

Substituting $n = 1$ into the floor expression gives -1 , and substituting $n = 2$ gives 0. Neither is the desired value 1. A uniform version valid for every positive n is

$$a(n) = \left\lfloor \frac{n^2 - 2}{3} \right\rfloor + 2\mathbf{1}_{\{n=1\}} + \mathbf{1}_{\{n=2\}}. \quad (7.3)$$

Equivalently,

$$a(n) = \frac{n^2}{3} - \frac{4}{3} + \frac{1}{3}\mathbf{1}_{\{3|n\}} + 2\mathbf{1}_{\{n=1\}} + \mathbf{1}_{\{n=2\}}. \quad (7.4)$$

Thus a second, coefficient-based derivation of the generating function is

$$A(z) = 2z + z^2 + \frac{z(1+z)}{3(1-z)^3} - \frac{4z}{3(1-z)} + \frac{z^3}{3(1-z^3)}. \quad (7.5)$$

Here the finite correction $2z + z^2$ accounts exactly for the two exceptional initial terms. Simplifying (7.5) gives the same rational function as (1.5).

8 Worked examples and the first cross-depth collision

8.1 Five copies of x

For $n = 5$, the classification is small enough to display completely.

Left depth k	Allowed s	$b = F'''(1)/3$	$F'''(1)$
1	1, 2, 3	3, 5, 7	9, 15, 21
2	1, 2	6, 8	18, 24
3	1	11	33
4	0	16	48

The resulting set is

$$\mathcal{D}_5 = \{9, 15, 18, 21, 24, 33, 48\},$$

so $a(5) = 7$. Within each row the second derivative is $2k$. The accompanying file `witnesses_n5_n7.json` gives an explicit fully parenthesized expression for every displayed joint state.

8.2 Seven copies of x

At $n = 7$, the normalized progressions and singleton are

k	Normalized third derivatives
1	3, 5, 7, 9, 11
2	6, 8, 10, 12
3	11, 13, 15
4	18, 20
5	27
6	36

There are 16 joint states, but the value $b = 11$ appears at both $k = 1$ and $k = 3$. Hence there are only 15 distinct third derivatives. The two expressions

$$F_1 = x^{C_6}, \quad F_2 = ((x^{R_4})^x)^x$$

make the collision explicit. For F_1 , $(k, s) = (1, 5)$, while for F_2 , $(k, s) = (3, 1)$. Both have $b = k^2 + 2s = 11$, so

$$F_1'''(1) = F_2'''(1) = 33, \quad F_1''(1) = 2, \quad F_2''(1) = 6.$$

They are therefore not the same analytic function, despite having the same third derivative at 1.

For $n \leq 6$, every term in the overlap correction is zero. Thus $n = 7$ is the first collision *between distinct joint derivative states*. Different parenthesizations can already have identical joint states at smaller sizes; these are a different kind of duplication.

8.3 Initial values and a counting audit

n	Joint states $1 + \binom{n-1}{2}$	Overlap correction	$a(n)$
1	1	0	1
2	1	0	1
3	2	0	2
4	4	0	4
5	7	0	7
6	11	0	11
7	16	1	15
8	22	2	20
9	29	3	26
10	37	5	32
11	46	7	39
12	56	9	47
13	67	12	55
14	79	15	64
15	92	18	74

The archive includes this audit through $n = 60$ and the sequence coefficients through $n = 1000$.

9 Consequences of the classification

9.1 Extremal derivatives

Corollary 9.1. For $n \geq 3$,

$$\min \mathcal{D}_n = 9, \quad \max \mathcal{D}_n = 3(n-1)^2.$$

The maximum is attained only by the fully left-associated expression C_n . For $n = 1, 2$, the derivative sets are $\{0\}$ and $\{3\}$, respectively.

Proof. In a non-extremal slice, $k \geq 1$ and $s \geq 1$, so $k^2 + 2s \geq 3$, with equality attained at $k = s = 1$. The exceptional singleton is larger than every other normalized value, as shown in [section 5](#). Attaining it forces $k = n - 1$, so every right subtree along the spine is a leaf. That is exactly C_n . $\square$

For $n \geq 3$, the minimum can be described just as explicitly: it is attained exactly by expressions $x^{(x^H)}$ with $H \in \mathcal{E}_{n-2}$. Indeed, $k = 1$ means $F = x^G$, and $s = 1$ means that G has left depth 1, hence $G = x^H$.

9.2 Simple recurrences

From the closed formula, for $n \geq 3$,

$$a(n+3) = a(n) + 2n + 3. \tag{9.1}$$

This follows because adding 3 to n adds the integer $2n + 3$ to $(n^2 - 2)/3$, so it adds the same integer to its floor. The forward differences are

$$a(n+1) - a(n) = \left\lfloor \frac{2n}{3} \right\rfloor \quad (n \geq 3), \quad (9.2)$$

as is checked directly in the three residue classes in (7.2). The stated starting indices matter: these recurrences do not extend unchanged across both exceptional initial terms.

Expanding the denominator in (1.4) also gives the homogeneous recurrence

$$\begin{aligned} a(n) = & 2a(n-1) - a(n-2) + a(n-3) \\ & - 2a(n-4) + a(n-5), \quad n \geq 8. \end{aligned} \quad (9.3)$$

The initial data are $a(1), \dots, a(7) = 1, 1, 2, 4, 7, 11, 15$. The numerator has degree 7, which explains the onset $n = 8$.

9.3 Quadratic growth and exact periodic correction

For $n \geq 3$, (7.4) gives the exact identity

$$a(n) = \frac{n^2}{3} - \frac{4}{3} + \frac{1}{3} \mathbf{1}_{\{3|n\}}. \quad (9.4)$$

In particular,

$$a(n) = \frac{n^2}{3} + O(1), \quad a(n) \sim \frac{n^2}{3}.$$

The constant correction is -1 on multiples of 3 and $-4/3$ on the other two residue classes. Thus the sequence is eventually a quadratic quasipolynomial of period 3; it is not eventually a single polynomial.

9.4 Why the denominator has degree five

The denominator factors as

$$(1-z)^2(1-z^3) = (1-z)^3(1+z+z^2).$$

The cubic factor at $z = 1$ corresponds to quadratic growth. The two primitive cube roots of unity encode the periodic correction. These factors do not cancel. For

$$N(z) = z(1 - z + z^2 + 2z^4 - z^6),$$

we have $N(1) = 2$. If $\omega^3 = 1$ and $\omega \neq 1$, then $N(\omega) = -\omega \neq 0$.

Consequently, (9.3) has the smallest possible order among eventual homogeneous recurrences with constant coefficients. Indeed, any such recurrence would give a polynomial $Q(z)$ of the same order for which $Q(z)A(z)$ is a polynomial. The reduced denominator of $A(z)$ must divide $Q(z)$, so $\deg Q \geq 5$. The shorter relation (9.1) is inhomogeneous and therefore does not contradict this statement.

10 Exact computation and reproducibility

The proof above is valid for every n and does not rely on finite verification. The archive nevertheless supplies independent exact checks, both to make the result reproducible and to catch implementation or indexing errors.

10.1 Three independent routes to the data

The companion `verify.py` uses only Python's standard library. It implements the following mathematically distinct computations.

Recursive derivative-state enumeration. Starting from $\{(0,0)\}$ at size 1, it combines all splits $n = j + (n - j)$ using (2.5). This computation does not use the claimed explicit state set or count. Because the right child's b -coordinate is irrelevant, its actually computed depth projection is used to avoid redundant work.

Exhaustive expression enumeration with Taylor arithmetic. All ordered binary parenthesizations are enumerated up to a selected small size. Each is evaluated in $\mathbb{Q}[t]/(t^4)$ after replacing x by $1 + t$, using

$$G^H = \exp(H \log G), \quad \log(1 + w) = w - \frac{w^2}{2} + \frac{w^3}{3}, \quad e^w = 1 + w + \frac{w^2}{2} + \frac{w^3}{6} \pmod{t^4}.$$

This evaluator does not use the derivative-state recurrence. If the resulting jet is $1 + t + c_2 t^2 + c_3 t^3$, then $(k, b) = (c_2, 2c_3)$. The coefficients are exact rational numbers, not finite-difference approximations.

Expansion of the proposed rational function. The script separately expands the original OEIS numerator and denominator by coefficient division. Its coefficients are checked against both the floor formula and the overlap formula. This route is also used to check (9.1) and (9.3).

The script additionally constructs a witness for every admissible joint state up to a selected size, counts its leaves, and checks its Taylor jet with the independent evaluator.

10.2 Executed checks

The supplied `verification_report.json` records a successful run with the following bounds:

Check	Executed bound
Recursive states versus the complete classification	$1 \leq n \leq 60$
All parenthesizations, evaluated by exact Taylor jets	$1 \leq n \leq 10$
Constructive witnesses, each checked by Taylor jets	$1 \leq n \leq 20$
Rational g.f. coefficients versus both count formulas	$1 \leq n \leq 1000$

There were 4,862 parenthesizations at $n = 10$ and 6,918 in total through $n = 10$. All 1,160 constructed witnesses through $n = 20$ passed their leaf-count and derivative checks. No floating-point differentiation or numerical tolerance was used.

For context, if $c_n = |\mathcal{E}_n|$, the recursive syntax gives $c_1 = 1$ and $c_n = \sum_{j=1}^{n-1} c_j c_{n-j}$. Hence $C(z) = \sum_{n \geq 1} c_n z^n$ satisfies $C = z + C^2$, and so $c_n = \binom{2n-2}{n-1}/n$. The exponential number of expressions explains why direct symbolic enumeration is useful for small independent checks but not needed for the proof or for large n .

10.3 How to reproduce the files

From the archive directory, run

```
python3 verify.py --out-dir artifacts
pdflatex -interaction=nonstopmode -halt-on-error article.tex
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```

The Python command has no third-party dependencies. Use Python 3.10 or later, without the `-O` option, since verification uses assertions. The PDF is built with a standard TeX Live or MiKTeX installation that includes the packages declared at the top of `article.tex`. A Makefile and detailed `README.md` are also supplied.

For a single n , the full state set is available by the short formula

```
states = {(n - 1, (n - 1)**2)} | {
    (k, k*k + 2*s)
    for k in range(1, n - 1)
    for s in range(1, n - k)
}
third_derivatives = sorted({3*b for k, b in states})
```

This formula also handles $n = 1$ and $n = 2$ correctly. It uses $O(n^2)$ state insertions for one size. By contrast, evaluating the count itself needs only a constant number of integer arithmetic operations via (1.2); this is not a claim of constant bit complexity for unbounded integers.

11 Scope of the result

The original task is resolved completely: (1.5) is proved for the sequence as defined, including its offset and initial terms. The stronger classification (4.1) supplies explicit witnesses and rules out missing derivative values. The argument counts distinct values, not their frequencies among parenthesizations.

Higher derivative counts form the broader table recorded as OEIS A216368 [2]. The present proof makes no assertion that the fourth- or higher-derivative sequences have analogous rational generating functions. The fact that only two integer statistics are needed here is a special feature of the third-order calculation.

A A second derivation of the spine formula

There is a useful analytic check on [lemma 3.1](#) that bypasses successive use of the derivative recurrence. On $(0, \infty)$, the ordinary identity $(u^v)^w = u^{vw}$ allows (3.1) to be written

$$F(x) = x^{P(x)}, \quad P(x) = \prod_{i=1}^k H_i(x).$$

At $x = 1$, each H_i and H'_i equals 1. With $s = \sum_i k(H_i)$, the product rule gives

$$P(1) = 1, \quad P'(1) = k, \quad P''(1) = \sum_i H''_i(1) + \sum_{i \neq j} H'_i(1)H'_j(1) = 2s + k(k-1).$$

Set $V = P \log x$. Then

$$\begin{aligned} V'(1) &= 1, \\ V''(1) &= 2k - 1, \\ V'''(1) &= 3P''(1) - 3P'(1) + 2 = 6s + 3k^2 - 6k + 2. \end{aligned}$$

Since $F = e^V$, we obtain

$$F''(1) = V''(1) + (V'(1))^2 = 2k$$

and

$$F'''(1) = V'''(1) + 3V'(1)V''(1) + (V'(1))^3 = 3k^2 + 6s.$$

The cancellation of the linear and constant terms makes the square k^2 especially transparent in this form.

B A compact proof suitable for a sequence note

For each parenthesized expression F , define $k(F) = F''(1)/2$ and $b(F) = F'''(1)/3$. Direct differentiation gives

$$k(G^H) = k(G) + 1, \quad b(G^H) = b(G) + 2k(G) + 2k(H) + 1,$$

with $k(x) = b(x) = 0$. Along a left spine of length k with right subexpressions H_i , this gives $b = k^2 + 2 \sum_i k(H_i)$. For size n , the resulting joint states are exactly

$$(n-1, (n-1)^2), \quad (k, k^2 + 2s) \quad (1 \leq k \leq n-2, 1 \leq s \leq n-k-1).$$

For realizability, put all $n-k$ nontrivial right-subtree leaves into one subtree of depth s and make the remaining right subtrees leaves; a size- m tree of any left depth $1, \dots, m-1$ is obtained by successively adding left-spine nodes to a right comb.

At fixed k , the possible b form a step-2 interval with endpoints $k^2 + 2$ and $(k-1)^2 + 2n - 3$. Only equal parities overlap. On inserting the k th interval, its intersection with the preceding union is exactly its intersection with the $(k-2)$ nd interval, of size $(n-3k+3)_+$. The extremal singleton is disjoint. Therefore

$$a(n) = 1 + \binom{n-1}{2} - \sum_{j \geq 2} (n-3j)_+.$$

Summing over $n \geq 1$ gives

$$\sum_{n \geq 1} a(n)z^n = \frac{z}{1-z} + \frac{z^3}{(1-z)^3} - \frac{z^7}{(1-z)^2(1-z^3)},$$

which simplifies to the conjectured rational function. Evaluating the finite sum gives $a(n) = \lfloor (n^2 - 2)/3 \rfloor$ for $n \geq 3$, with $a(1) = a(2) = 1$.

References

- [1] The OEIS Foundation Inc., A199085: *Number of distinct values taken by the third derivative of parenthesized self-exponentials at $x = 1$* , <https://oeis.org/A199085>. Text endpoint supplied with the problem: <https://oeis.org/search?q=id:A199085&fmt=text>. Accessed September 20, 2026.
- [2] The OEIS Foundation Inc., A216368: *Counts of distinct derivative values of parenthesized self-exponentials*, <https://oeis.org/A216368>. Accessed September 20, 2026.